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DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING



Network Survey Report on

"NETWORK TOPOLOGY ARCHITECTURE IN BMSIT&M"

BACHELOR OF ENGINEERING
in
INFORMATION SCIENCE AND ENGINEERING
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ABSTRACT

Network architecture with its security is a growing concern over the recent years. A campus network faces challenges to address core issues of security which are governed by the network architecture. This report is mainly targeted towards BMSIT&M campus network which delivers the required security. This is essential because it prevents the institution from suffering any significant attacks associated with the network.

A university network has a number of uses such as teaching, learning, research, management, e-library, and connections with the external resources. Therefore, network architecture and its security are vital issues for any university. Network security isn't just an option, it's a necessity. Students now regularly have access to smartphones and tablets because online access has turned into a crucial tool for learning. To make sure important campus data and files are protected, administrators need a stable and secure environment, which is where network security comes in. A well-managed network security system will feature measures where administrators can efficiently control and monitor what students and teachers access while online to prevent a breach in the network. Web filters can certainly play a role in restricting certain websites, network security can go much further by taking a more proactive approach in monitoring online activity and blocking sites that may lead to security compromises.

ACKNOWLEDGEMENT

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INTRODUCTION

Data centers are becoming increasingly popular for their flexibility and processing capabilities in the modern computing environment. They are managed by a single entity (administrator) and allow dynamic resource provisioning, performance optimization as well as efficient utilization of available resources. Each data center consists of massive storage resources connected with physical wires such as UTM, core switches, node switch, OFC, etc.

Unified Threat Management (UTM) provides multiple security features and services in a single device or service on the network, protecting users from security threats in a simplified way. UTM includes functions such as anti-virus, anti-spam, content filtering, and web filtering and the core switch is a high-capacity switch generally positioned within the backbone or physical core of a network.

Core switches serve as the gateway to a wide area network (WAN) or the Internet - they provide the final aggregation point for the network and allow multiple aggregation modules to work together.

Node switch is an electronic networking device that is attached to a network, and is capable of creating, receiving, or transmitting information over a communication channel.

Optical Fiber Cable (OFC) is one that has glass fibers which are inside an insulated casing. The purpose of a Optical Fiber Network is to reach longer distances and provide a higher performance than that of wired cables**.**
The large scale nature of data centers requires careful planning of compute, storage, network nodes, interconnection as well as inter-communication for their effective and efficient operations.

In this report, we present a comprehensive survey and taxonomy of network topologies used in our college data center.

Network Infrastructure

The Internet facility has been crucial in the evolution of our education system in many ways. In order to support its employees and students with best IT infrastructure, BMSIT&M has established one of its best, robust network infrastructure in 2019 - 2020 to facilitate reliable and high speed internet service to both Students, Faculty and Researchers at the campus and BMSIT&M Hostels.

BMSIT&M has set up its own **DATA CENTER** with all advanced equipment like UTM, Core switch, advanced WIFI Access Points (141nos) OFC cabling to interconnect buildings inside campus and RF technology to connect Ladies hostel too.

Customer Base and Authentication System:

Total users on board: 4500

Concurrent Users: 800-1000

Internet user Authentication for WiFi: Extreme NAC is a centralized authentication system server installed at data centres through which users are created and given to registered users through google form keeping the USN as a unique key for students and employee ID as key for staff.

Bandwidth per User: data per user and bandwidth of 100Mbps upper bandwidth is reserved per user. It is shared as the user increases in the network.

Bandwidth backhaul and distribution: Band width from two ISPs (350+200=550) Mbps of 1:1 Internet bandwidth Bandwidth is terminated at Data centre and being distributed to the entire campus through optical fiber cable.

O&M: Two dedicated technical staff to attend and address real time issues, operations and maintenance.

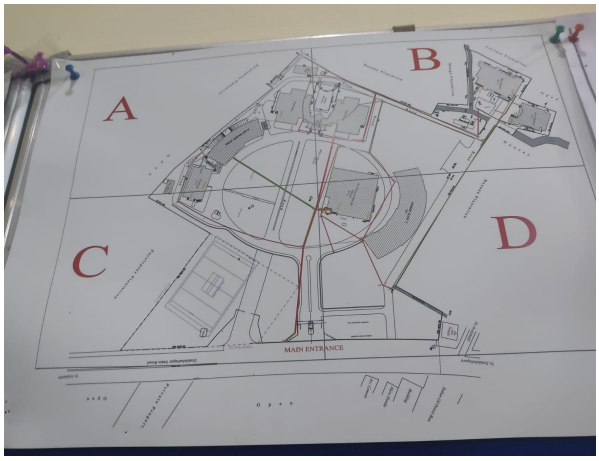
A third party vendor is also on call for any new or existing error resolution.

Main points observed:

1. Campus includes 5 blocks:
    - Academic block
    - Architecture block
    - BSN block
    - Canteen
    - Hostel
    - Lab block
2. All 6 blocks are connected using Star topology.
3. Within the block the switches are connected using cascading topology
4. BSNL and convergent are the service providers.
5. Both the service providers provide bandwidth of 40 mbps.
6. 40 mbps is divided into 5 categories:
    - Top users
    - Students
    - Staff
    - Office
    - Placement cell
7. The above packet division and database management is done using 24 online client servers.

Structure of the Network Architecture

The network architecture that has been adopted in BMSIT&M college is a **STAR based Hybrid Topology**, which has come to be one of mostly widely used in the recent days. In the campus, the Hub originates from the Data Center of the college and is drawn to respective buildings.



Network Distribution

- Individual requests from Wifi/LAN network connected devices reach the floor switch.
- Multiple floor switches connect together at a common node for the block. In fact, BSN block has 2 block nodes (switches) for convenience purposes.



- All of these in turn are connected to their respective stacks as shown in the image here.
- These stacked nodes are connected to the core switch (purple in image) by using patch cords.
- Usually there is only one core switch required, but in BMSIT we use two core switches.

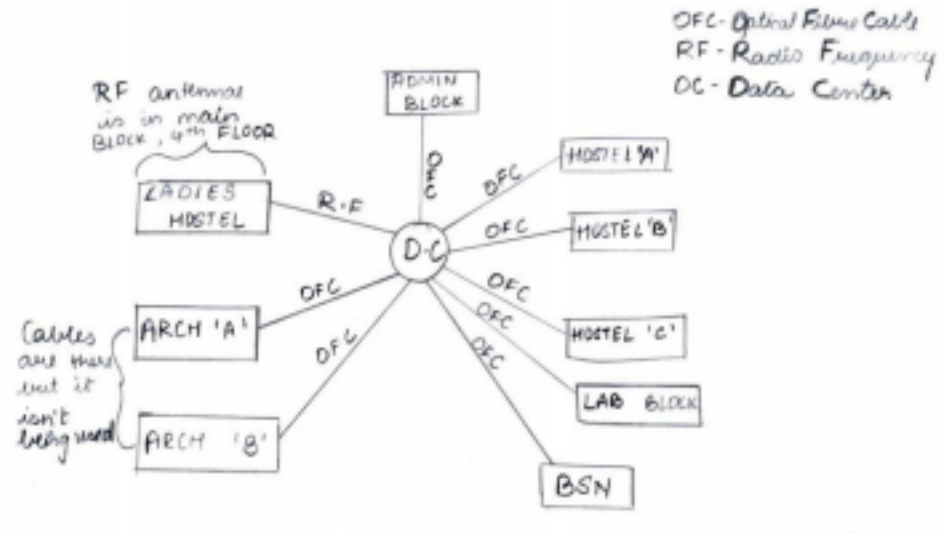


- From the core switch(es), the data enters the UTM (Unified Threat Management) system, where restricted sites are blocked, browsing policies take effect, and cyber attacks are prevented.
- The next component are the network connectivity modules from Tata Teleservices and DVOIS where internet access happens, according to the aforementioned bandwidths.



Topology adopted by College

STAR based Hybrid Topology is implemented in the college.



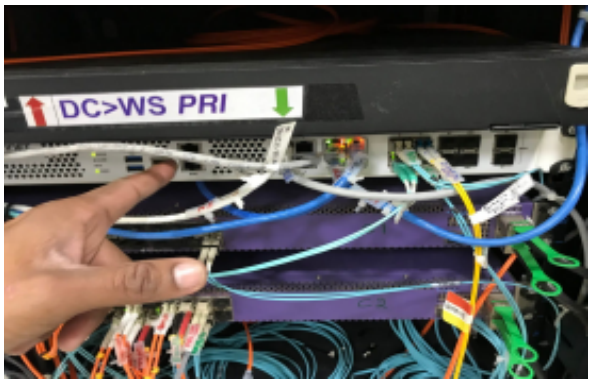
Difficulty in setting up a Network in the college

- Challenging activities in general.
- Where's the data center(accessible location)
- How to lay cables
    - 8ft down
    - Costly
    - Trenchless digging
    - Drilling pipes contain HDPE which contains OFC.



Services to other Branches:

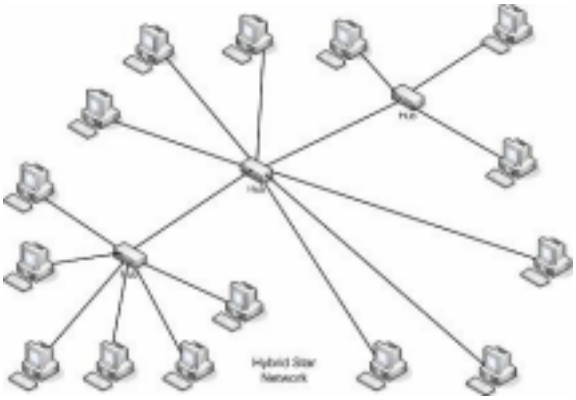
There are some services provided to the Girls Hostel:



- The RF (Radio Frequency) Wireless Antenna is located on the 4th floor of the Admin Block and ,one at the Girls Hostel.
- There is a Mutual IP for communication between the Girls Hostel and the Data Center as the Girls Hostel will have the Data Center IP and the Data Center will have Girls Hostel IP.

What made them choose Star Topology?

- Isolation of devices
- Simplicity
- If any cable is not working then the whole
  network will not be affected
- You can easily add new computers or devices
  to the network without interrupting other nodes
- Centralization
- Easy to troubleshoot
- Better performance
- Easy Installation

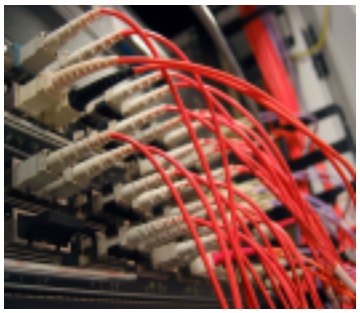


Network Systems Used:

They have used both wired and wireless network systems:

- Wired
    - OFC(Optical Fiber Control)
- Wireless
    - RF for Girls Hostel
    - Wifi





Bandwidth on the campus network:

550MBPS (350 Primary + 100 Secondary (backup) DVOIS + 200 from Tata)



Internet Service Provider:

DVOIS

TATA TELESERVICES

College Network Layout:

The BMSIT&M campus is divided into 4 quadrants:

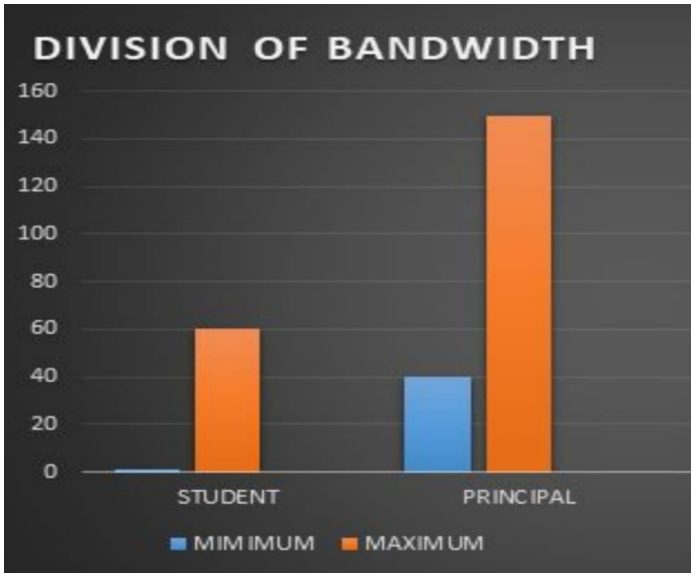
- A - Architecture Block and Left Side of the Main Block
- B - Right Side of Main Block and the Boys Hostel
- C - Main Entrance
- D - BSN Block and the Lab Block



The "**RED LINE**" represents the **PRIMARY OFC** which connects all the 4 quadrants to the DATA CENTRE for internet access.
The "**GREEN LINE**" represents the **SECONDARY OFC** which is used as backup in case of an internet blackout.
The "**YELLOW DOT**" represents the **DATA CENTRE** located in the BSN Block.

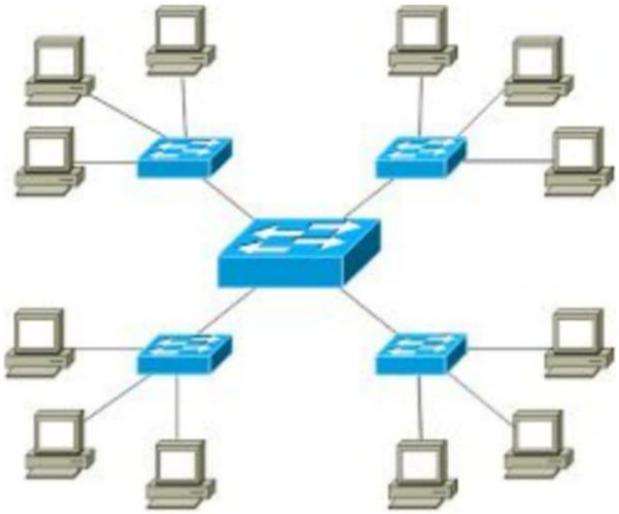
Division of the Bandwidth:

- Students- min 1MBPS, 60MBPS.
- Principal-min 40 MBPS, max 150 MBPS at times.



Alternatives used in case of unstable connection to the internet :

There are secondary lines, like a ring topology with intermediate connections to the data center which helps as an alternative.



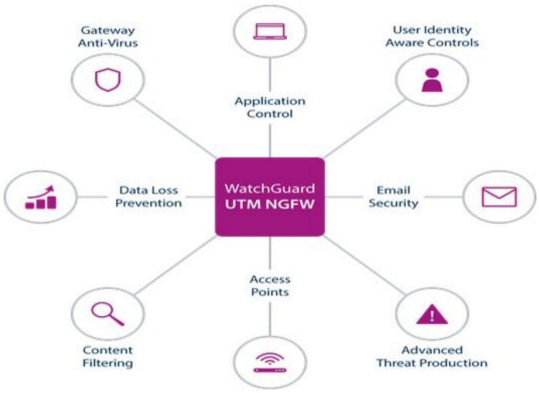
IP addresses divided between the different blocks:

- There is only one internal IP.
- **0** series belongs to admins, **1** series to the departments.
- Wifi in separate series.
- Two lines come to the core switch of which, one is wifi and the other is LAN and both have separate IP addresses.
- From which it goes to the UTM.
- Based on the session, bandwidth depends.
- First priority is given to LAN.

What precautions are taken against cyber attacks?

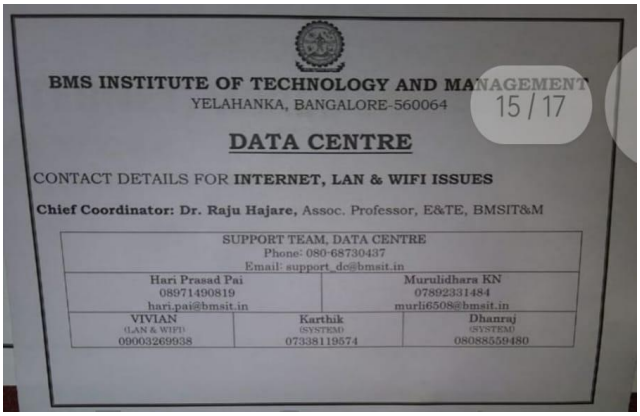
- UTM(Unified Threat Management System) is used.
- Stay ahead by backing up data and storing it separately.
- Educate staff on the dangers of unsecured networks.
- Use robust anti-malware and firewall software.





Network Administrator:

No particular person. A group of people do it. There's a coordinator, a bunch of people and Support staff as well.



What cybersecurity tools are available?

The tools available is UTM which has many advantages such as:

- They are flexible and they can be easily adapted.
- They have a centralized integration and management system.
- They are cost effective.
- They have increased awareness of network security threats.
- They have a faster security solution.

Conclusion

This network architecture survey was primarily aimed at understanding the topology and the types of networking devices that have been used on the campus. We believe that this survey helped us understand and comprehend how important network security is to an organization and we would like to use this knowledge gained through this survey and follow all the norms and protocols to prevent any kind of data breach.

Finally, we would like to thank Prof. Gireesh Babu and Mr. Muralidhara for providing us the opportunity to take up this survey and in helping us gather the required information.

RESOURCES

- https://www.researchgate.net/publication/242077540_Design_and_Implementation_of_Network_Topology_Discovering_System
- https://arxiv.org/abs/1605.01701
- https://bmsit.ac.in/network-infrastructure
- https://www.juniper.net/us/en/products-services/what-is/utm/
- https://www.techopedia.com/definition/28787/core-switch-networking#:~:text=A%20core%20switch%20is%20a,aggregation%20modules%20to%20work%20together
- https://www.google.com/imgres?imgurl=https%3A%2F%2Fsnabaynetworking.com%2Fwp-content%2Fuploads%2F2019%2F11%2FStar-Topology-Advantages-and-Disadvantages.jpg&imgrefurl=https%3A%2F%2Fsnabaynetworking.com%2Fstar-topology%2F&tbnid=49lAfm-P6agkgM&vet=12ahUKEwi425uhku7tAhXInEsFHRzaDPEQMyhQegUIARCdAQ..i&docid=ggC_z0ppnZLrNM&w=700&h=400&q=%20star%20topology&ved=2ahUKEwi425uhku7tAhXInEsFHRzaDPEQMyhQegUIARCdAQ
- https://docs.google.com/presentation/d/1pbRD2sEY72Mi0YUR_sNB_LV--USh5swsWPTlVceyJv4/edit?ts=5fe84805#slide=id.g789ccd6e34_0_25